

ZS-F35

The Multiplicity-Free Duality-Singlet Theorem and the Structural Dimensionless Factor of the Z-Spin Vacuum Susceptibility

Author: Kenny Kang

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Theme / Code: Foundations / ZS-F35

Version: 1.5 (terminal; supersedes v1.4, v1.3, v1.2, v1.1, v1.0)

Verification: 23 PROVEN + 4 IDENTITY PASS | 6 ASSUMPTIONS tracked (not derivations) | general theorem T1 (proof = Schur), Lorentzian bridge, and the FULLY EXACT character-field obstruction T2 — symbolic commutant (rank 34, nullity 2), $s^2 = 5I$, $\det s = -125$, $[s, R(g)] = 0$ for all $60\ g \in A_5$, $\star = (1/\sqrt{5})s$ non-rational — machine-checked in sympy | structural dimensionless factor derived; one UV-normalization gate (C_norm) left to ZS-F36 | Zero fitted parameters | $(A, Q, \dim Z) = (35/437, 11, 2)$ LOCKED.

§0. Abstract

We prove a general representation-theoretic result and a field-of-definition obstruction, and use them to isolate and derive the **structural dimensionless factor** of the Z-Spin odd-sector vacuum susceptibility, leaving one explicit UV-normalization gate.

Multiplicity-Free Duality-Singlet Theorem (T1). Let a finite group G act on a finite-dimensional unitary complex representation $V = U_+ \oplus U_-$, where U_+ , U_- are non-isomorphic absolutely irreducible G -modules of equal complex dimension d , each of multiplicity one. Then the commutant is $\text{End}_G(V) = \mathbb{C}P_+ \oplus \mathbb{C}P_-$; the unique traceless invariant (up to scale) is the duality involution $J = P_+ - P_-$, with $J^2 = I$, $\text{Tr } J = 0$, and the Hilbert–Schmidt norm (relative to the invariant Hermitian inner product) $\|J\|_{\text{HS}}^2 = \text{Tr}(J^\dagger J) = 2d$. If a CP-odd source couples as $\hat{c}_\theta = \lambda J$, then with the order-parameter metric G (with $G(J, J) = G_J \|J\|_{\text{HS}}^2$) the response is $\hat{c}_\theta^\dagger G^{-1} \hat{c}_\theta = |\lambda|^2 \cdot (2d)/G_J$, a single scalar. The invariant $\|J\|_{\text{HS}}^2 = 2d$ is carrier-independent; $J^2 = I$ additionally requires $V = U_+ \oplus U_-$ exactly. The theorem follows from Schur’s lemma.

Z-Spin instance. With $G = A_5$ and the rank-6 bivector carrier $\Lambda^2 V_4 = \mathbf{3} \oplus \mathbf{3}'$ (F34.BIV), the duality involution is the Hodge star $\star$, and $v_{s^2} = \|\star\|_{\text{HS}}^2 = 2d = 6 = \dim Y$. The source is $\star$ on **complementary grounds**: a representation-theoretic uniqueness (T1) and a physical Pontryagin-source identification ($\delta S_\theta / \delta \theta$ has operator $\star$ since $F \wedge F = \langle F, \star F \rangle$). The v1.1 group-theory error is corrected (the central inversion of $I_h \cong A_5 \times \mathbb{Z}_2$ cannot swap $3 \leftrightarrow 3'$; the swap is the outer automorphism, $S_5 \not\cong A_5 \times \mathbb{Z}_2$, physical realization OPEN = F34 G-Outer-Physical). The Euclidean computation transfers to Lorentzian signature via $J_L = -i\star_L$ ($J_L^2 = I$).

Character-field obstruction (T2). The endomorphism algebra of the rational form is $\text{End}_{\{\mathbb{Q}A_5\}}(\mathbf{3} \oplus \mathbf{3}') = \mathbb{Q}(\sqrt{5})$ — generated over $\mathbb{Q}$ by an integer operator s with $s^2 = 5I$, $\text{Tr } s = 0$, $\det s = -125$, commuting with the entire A_5 action — and the duality involution $\star = (1/\sqrt{5})s$

requires the irrational coefficient $1/\sqrt{5}$. Hence no A_5 -stable $\mathbb{Q}$ -form (a fortiori no A_5 -stable $\mathbb{Z}$ -lattice) makes $\star$ rational; the $\text{disc}(A_4) = 5$ root lattice is one concrete instance. This is a special case of a general field-of-definition phenomenon: for any Galois-conjugate pair of absolutely irreducible modules with a real quadratic character field $\mathbb{Q}(\sqrt{m})$, the duality involution is $(1/\sqrt{m})s$ with $s^2 = mI$, never rational. (Verified fully exactly in $\mathbb{Q}(\sqrt{5})$: symbolic commutant, rank 34 / nullity 2, $s^2 = 5I$, $[s, R(g)] = 0$ for all $g \in A_5$.) Consequently $v_{s^2} = 6$ is the Hilbert–Schmidt norm relevant to the kinetic susceptibility; flux quantization belongs separately to the integral cohomology lattice, which the Hodge involution need not preserve. T2 identifies this metric–arithmetic mismatch — the icosahedral $\sqrt{5}$ /golden ratio.

Result. Collapsing the convention-dependent normalizations into one observable stiffness $\tilde{G}_s := V_\Sigma G_s / Z_{\text{match}}$, the dimensionless coefficient factorizes:

$$\chi^{(s)} = (\dim Y)^2 \cdot (A/Q) \cdot C_{\text{norm}} \cdot M_{UV}^4 = (1260/4807) \cdot C_{\text{norm}} \cdot M_{UV}^4, C_{\text{norm}} = \tilde{G}_s^{-1} \cdot (c_e/2\pi)^2.$$

The arithmetic identity $36 A/Q = 1260/4807$ is PROVEN; that this is the physical structural factor is DERIVED-CONDITIONAL on $\hat{c}_\theta = \star$ (the F34.BIV carrier reduction). Under the Canonical UV Normalization $C_{UV} = \{\tilde{G}_s = 1, c_e = 2\pi\}$, $C_{\text{norm}} = 1$ exactly, giving $\chi^{(s)} = (1260/4807) M_{UV}^4 \approx 0.262 M_{UV}^4$; but $C_{\text{norm}} = 1$ is the closure gate deferred to ZS-F36. The residual is **one dimensionless gate (C_{norm}) plus one irreducible dimensionful scale (M_{UV})**. Verification: 23 PROVEN + 4 IDENTITY PASS (NumPy + SymPy).

Epistemic Status Legend

Tag	Meaning
PROVEN	Explicit proof or exact machine verification; no undischarged assumption.
DERIVED	Follows from PROVEN results by stated steps; no new parameter.
DERIVED-CONDITIONAL	Derived modulo explicitly named, falsifiable conditions.
DERIVED-BY-INHERITANCE	Uses an upstream corpus result, not re-proven here.
HYPOTHESIS-strong	Structurally motivated; a key value/identification not yet proven.
NO-GO	A proven impossibility/non-uniqueness disciplining the program.
IDENTITY / REGRESSION	An arithmetic or cross-version consistency; no new physical content.
ASSUMPTION-CONSISTENCY	A check that a stated assumption is internally consistent — not its derivation.
OPEN	Conceptually unresolved; a genuine gap.
PROVEN-irreducible	Proven that the quantity cannot be fixed by the stated inputs.

§1. Introduction

ZS-F34 reduced the odd-sector vacuum susceptibility to $\chi_- = (Z_{\text{match}} g_{\text{reg}}^2 / 4\pi^2 V_{\Sigma}) e_6^2 \hat{c}_{\theta^T G^{-1} \hat{c}_{\theta}}$ and called itself a *reduction, not a closure*. This paper's history is a record of progressive correction: v1.0 rearranged unknowns; v1.1 over-claimed a closure on a flawed group-theoretic step; v1.2 reframed the result as a conditional factorization; v1.3 corrected the C_{norm} definition, strengthened the verification, and lifted the content to a general theorem and a lattice question. A fourth review identified four remaining issues — an over-broad theorem hypothesis, a verify/probe inconsistency, a T2 conclusion stronger than its computation, and a missing Lorentzian bridge — and a route to make the lattice obstruction a genuine theorem. v1.4 implements all of them and is the **terminal** version of ZS-F35.

The central content is a general theorem (T1, §3): in a finite-symmetry theory, a multiplicity-free pair of equal-dimension absolutely-irreducible modules carries a unique traceless invariant — a duality involution — to which a CP-odd source couples, reducing the susceptibility to a single scalar. Z-Spin is the case $G = A_5$, $d = 3$, $J = \star$ (§4). The norm $v_{\text{s}}^2 = 2d = 6$ is the Hilbert–Schmidt norm; a field-of-definition theorem (T2, §5) proves the duality involution is non-rational on every A_5 -stable $\mathbb{Q}$ -form, with $\text{disc}(A_4) = 5$ — the icosahedral $\sqrt{5}$ — as one lattice instance. The remaining normalizations collapse into one observable stiffness and one Dirac coefficient (§§6–7); the result is the factorization above, with one dimensionless gate and one dimensionful scale left for ZS-F36 (§10).

§2. The cosmic split and the single invariant mode

The register channel algebra is $\text{End}(V_{11})$, $V_{11} = \mathbb{C}^{12} \oplus 1$, character $(11, -1, -1, 1, 1) = 3 \oplus 3' \oplus 5$.

As an A_5 -module,

$$\text{End}(V_{11}) = 3 \cdot 1 \oplus 6 \cdot 3 \oplus 6 \cdot 3' \oplus 8 \cdot 4 \oplus 10 \cdot 5, \dim = 121 = Q^2.$$

The trivial isotypic is three-dimensional. **[PROVEN P1.]** The arithmetic split $6 + 32 + 83 = 121$ with three trivial copies is PROVEN (P2a); the physical assignment of dim 6 to the baryon bivector $3 \oplus 3'$, 32 to CDM, and 83 to dark energy, with two trivial copies in CDM and one in dark energy, is DERIVED-BY-INHERITANCE from the upstream cosmic split (F34/A19).

[P2b.] Within the exact- A_5 , invariant-source branch (one of the three source branches of F34.SR), the dark-energy effective mode count is $N_{\text{eff}} = 1$.

§3. The Multiplicity-Free Duality-Singlet Theorem (general)

Theorem T1. Let G be a finite group and $V = U_+ \oplus U_-$ a finite-dimensional unitary complex representation, where U_+ , U_- are non-isomorphic **absolutely irreducible** G -modules with $\dim U_+ = \dim U_- = d$, each of multiplicity one. Let P_+ , P_- be the equivariant orthogonal projectors (with respect to a chosen G -invariant Hermitian inner product). Then:

- (i) **(Schur)** $\text{End}_G(V) = \mathbb{C}P_+ \oplus \mathbb{C}P_-$, of complex dimension 2.
- (ii) The trace map on the two-dimensional commutant has one-dimensional kernel $\mathbb{C}(P_+ - P_-)$; the **duality involution** $J := P_+ - P_-$ spans it, and the identity $I = P_+ + P_-$ provides a canonical complementary line.

- (iii) $J^2 = I$, $\text{Tr } J = 0$, and the **Hilbert–Schmidt norm** relative to the invariant Hermitian structure is $\|J\|_{\text{HS}}^2 = \text{Tr}(J^\dagger J) = \dim U_+ + \dim U_- = 2d$. The invariant $\|J\|^2 = 2d$ is *carrier-independent*; $J^2 = I$ additionally requires $V = U_+ \oplus U_-$ exactly.
- (iv) If a CP-odd (θ -type) source couples as $\hat{c}_\theta = \lambda J$ ($\lambda \in \mathbb{C}$), let G be the metric on the order-parameter space with $G(J, J) = G_J \|J\|_{\text{HS}}^2$ (G_J the stiffness in the J direction). Then $\hat{c}_\theta^\dagger G^{-1} \hat{c}_\theta = |\lambda|^2 \cdot (2d)/G_J$ — a single scalar.

Proof. (i) Over $\mathbb{C}$ with $U_\pm$ absolutely irreducible, Schur’s lemma gives $\text{Hom}_G(U_+, U_-) = \text{Hom}_G(U_-, U_+) = 0$ and $\text{End}_G(U_\pm) = \mathbb{C}$; hence the commutant is $\mathbb{C}P_+ \oplus \mathbb{C}P_-$. (ii) $\text{Tr}(aP_+ + bP_-) = (a + b)d$, vanishing iff $a = -b$. (iii) $J = +1$ on U_+ , -1 on U_- , so $J^2 = I$ and $\text{Tr } J = 0$; $\|J\|_{\text{HS}}^2 = \text{Tr}(P_+) + \text{Tr}(P_-) = 2d$. The HS norm uses the invariant Hermitian structure (it is not invariant under non-orthogonal change of basis); $J^2 = I$ uses $P_+ + P_- = I_V$, valid only when V is exactly $U_+ \oplus U_-$. (iv) Apply F34.SR with $c = \lambda J$ and G^{-1} acting as G_J^{-1} on the J line.

The theorem **follows analytically from Schur’s lemma**; it is illustrated numerically for three groups — A_5 ($d = 3$), S_5 ($d = 4$), S_3 ($d = 1$) — and the carrier-dependence of $J^2 = I$ is exhibited explicitly ($\|J\|^2 = 2d$ on both an exact and an enlarged carrier, $J^2 = I$ only on the exact one).

[Illustrations T1-A5, T1-S5, T1-S3, T1-carrier, T1-chi.]

Balanced-involution converse (remark). A G -invariant involution J ($J^2 = I$) with $\text{Tr } J = 0$ on a $2d$ -dimensional carrier forces each ± 1 eigenspace to have dimension d ; if the two eigenspaces are non-isomorphic irreps (equivalently, exchanged by an outer automorphism), the carrier is exactly $U_+ \oplus U_-$. A *balanced* duality involution thus selects the $d \oplus d$ split structurally.

T1 is a direct corollary of Schur’s lemma; its value here is in combination with the Pontryagin source identification (§4.3) and the field-of-definition obstruction (§5).

§4. The Z-Spin instance ($d = 3$)

§4.1 Group-theory correction: $I_h \neq S_5$

The two A_5 -triplets $3, 3'$ are exchanged by the outer automorphism $\sigma \in \text{Out}(A_5)$, realized by an odd permutation in S_5 . The v1.1 claim that the full icosahedral symmetry $I_h \cong A_5 \times \mathbb{Z}_2$ forces this is **incorrect**: the $\mathbb{Z}_2$ of I_h is the central inversion, which commutes with A_5 and (Schur) acts as a scalar on each irrep, hence *preserves* $3, 3'$. The swap-implementer is non-central (**P4, P6**), belonging to $S_5 \setminus A_5$, and $S_5 \not\cong A_5 \times \mathbb{Z}_2$ (trivial center). The physical realization of σ is the OPEN gate **G-Outer-Physical** inherited from F34. A symmetry-based forcing is unavailable; §§4.2–4.3 are used instead.

§4.2 Carrier uniqueness, and the Lorentzian bridge

By Theorem T1 with $G = A_5$, $V = \Lambda^2 V_4 = \Lambda^2_+ \oplus \Lambda^2_- = 3 \oplus 3'$ (F34.BIV), $d = 3$: the A_5 -commutant is $\text{span}\{P_3, P_{3'}\}$ (dimension 2; **P5**), and the unique traceless invariant is $\star = P_3 - P_{3'}$. A character check confirms the eigenspaces are the irreps $3, 3'$ (χ at a 5-cycle equals $\bar{\phi}, \phi$; **P3**); the traceless commutant is exactly $\text{span}\{\star\}$ (**P7**, residual 10^{-16}).

Signature. The representation-theoretic computation is performed in Euclidean signature, where the 4-dimensional 2-form Hodge star satisfies $\star^2 = +1$, giving the real self-dual/anti-self-dual split $3 \oplus 3'$. In Lorentzian signature $(1, 3)$ the 2-form Hodge star satisfies $\star_{-L}^2 = -1$, with

eigenvalues $\pm i$ on the complexified bivector; the corresponding involution is $J_L := -i\star_L$, satisfying $J_L^2 = I$ with ± 1 eigenspaces (3 each). The algebraic identities ($\star_L^2 = -I$, $J_L^2 = I$) are PROVEN; the further step of applying the Euclidean $3 \oplus 3'$ susceptibility norm to the physical Lorentzian EFT uses this complexification/Wick-rotation identification and is DERIVED-CONDITIONAL. [PROVEN: L1, L2, L3.]

§4.3 The Pontryagin source: an action-level derivation

The CP-odd (θ) coupling of a 2-form field strength F is the topological term $S_\theta = (\theta/8\pi^2) \int \{M_4\} F \wedge F$. In four dimensions the Pontryagin density equals the Hodge pairing, $F \wedge F = \langle F, \star F \rangle \text{vol}$, so $\delta S_\theta / \delta \theta = (1/8\pi^2) \langle F, \star F \rangle$, and the θ -source operator is exactly $\star$. Two conditions complete the bridge: (a) the F34 odd three-form A_3^- reduces (on $Y_6 = M_4 \times \Sigma_2$) to this bivector Pontryagin source — the F34.BIV carrier reduction; (b) no other A_5 -invariant CP-odd operator mixes — *guaranteed by Theorem T1*, since $\star$ is the only traceless A_5 -invariant on the carrier. Hence $\hat{c}_\theta = \star$, giving $v_{s^2} = \|\star\|^2_{\text{HS}} = 2d = 6 = \dim Y$.

The two arguments are **complementary, not independent**: the mathematics (T1) supplies a unique traceless line; the physics (the Pontryagin action) supplies that the θ -source lies on that line, and its tracelessness. With (b) PROVEN via T1, $\hat{c}_\theta = \star$ is DERIVED modulo only the carrier reduction (a). [S1: DERIVED-CONDITIONAL on the F34.BIV carrier reduction.]

§4.4 Why A_5 -equivariance alone is insufficient (NO-GO)

In the full three-dimensional trivial isotypic of $\text{End}(V_{11})$ without the carrier restriction or coupling identification, the source would be a free line orthogonal to the CDM 2-plane, and v_{s^2} would not be invariant (3 vs 49 for two admissible embeddings). [PROVEN P10.] The carrier/coupling identification of §§4.2–4.3 is load-bearing.

§5. The character-field obstruction (T2)

Theorem T2 (no rational duality involution). Let $W = 3 \oplus 3'$ be the six-dimensional rational A_5 -representation (the Galois orbit of the two absolutely-irreducible triplets). Then the rational endomorphism algebra is the **character field**

$$\text{End}_{\mathbb{Q}}\{\mathbb{Q}A_5\}(W) = \mathbb{Q}(\sqrt{5}),$$

generated over $\mathbb{Q}$ by an **integer** operator s (in the A_4 root basis) with $s^2 = 5I$, $\text{Tr } s = 0$, $\det s = -125$, and $[s, R(g)] = 0$ for all $g \in A_5$. The duality involution $\star = P_3 - P_{3'}$ satisfies $\star = (1/\sqrt{5}) \cdot s$, whose coefficient $1/\sqrt{5} \notin \mathbb{Q}$. Therefore $\star$ is not a rational endomorphism of any A_5 -stable $\mathbb{Q}$ -form — equivalently, $\star$ does not preserve any A_5 -stable $\mathbb{Q}$ -lattice — and a fortiori it is non-integral on every A_5 -stable $\mathbb{Z}$ -lattice. [PROVEN fully exactly in $\mathbb{Q}(\sqrt{5})$ via a symbolic commutant (rank 34, nullity 2), with no kron/vec convention: T2-int, T2-dim, T2-exact, T2-comm, T2-obs.]

Proof. W is $\mathbb{Q}$ -irreducible (a single Galois orbit under $\sqrt{5} \mapsto -\sqrt{5}$, which exchanges 3 and 3'); its endomorphism algebra is the field generated by the character, $\mathbb{Q}(\chi_3) = \mathbb{Q}(\varphi) = \mathbb{Q}(\sqrt{5})$, of degree 2. A general rational endomorphism is $pI + qs$ ($p, q \in \mathbb{Q}$), acting as $p \pm q\sqrt{5}$ on 3, 3'. Requiring ± 1 (the eigenvalues of $\star$) forces $p = 0$, $q = 1/\sqrt{5} \notin \mathbb{Q}$. If instead $\star$ preserved an A_5 -stable $\mathbb{Q}$ -lattice, $\ker(\star - I)$ and $\ker(\star + I)$ would be 3-dimensional rational A_5 -submodules, i.e. 3 and 3' would each be defined over $\mathbb{Q}$ — impossible, since their characters are irrational.

The numerical $\text{disc}(A_4) = 5$ finding ($\star_{\text{root}} = (1/\sqrt{5}) \cdot M$ in the A_4 root lattice) is **one concrete lattice manifestation** of this obstruction; the integer M is the generator s , with $M^2 = 5I$. The integrality obstruction is exactly the icosahedral $\sqrt{5}$ / golden ratio.

General form (Galois-conjugate pairs). The argument is not special to A_5 . Let U, U^σ be a Galois-conjugate pair of absolutely irreducible G -modules whose common character field $K = \mathbb{Q}(\chi_U)$ is a real quadratic field $\mathbb{Q}(\sqrt{m})$ ($m > 0$ squarefree), with σ the nontrivial element of $\text{Gal}(K/\mathbb{Q})$ exchanging U and U^σ . Then $\text{End}_{\mathbb{Q}}\{U \oplus U^\sigma\} = K = \mathbb{Q}(\sqrt{m})$, generated over $\mathbb{Q}$ by an operator s — integral on a G -stable order — with $s^2 = mI$ and $\text{Tr } s = 0$, and the duality involution $\star = P_U - P_{U^\sigma} = (1/\sqrt{m})s$ requires the coefficient $1/\sqrt{m} \notin \mathbb{Q}$. Hence $\star$ is never rational on a G -stable $\mathbb{Q}$ -form. The Z-Spin case is $m = 5$ ($G = A_5$), with $\text{disc}(A_4) = 5$ as one realization; the same statement holds verbatim for any such pair, the coefficient $1/\sqrt{m}$ being the field-of-definition obstruction. This is the externally most transferable result of ZS-F35.

Interpretation (metric vs arithmetic). The value $v_{s^2} = 6$ is the **Hilbert–Schmidt norm** relevant to the kinetic susceptibility (the quadratic form $\hat{c}_\theta^\dagger G^{-1} \hat{c}_\theta$). Flux quantization belongs **separately** to the integral cohomology (charge) lattice, which is metric-independent and which the Hodge involution **need not preserve**; T2 identifies precisely this metric–arithmetic mismatch for the A_4 realization. $v_{s^2} = 6$ is therefore a kinetic-norm statement, not a claim about the charge lattice, and is unaffected by the obstruction.

§6. The single observable stiffness

ZS-F34 carried three convention-dependent normalizations — $Z_{\text{match}}, V_\Sigma, G_s$ — only in the combination $Z_{s^{\text{phys}}} = V_\Sigma G_s / (Z_{\text{match}} g_{\text{reg}}^2)$. Define the **single observable kinetic stiffness**

$\tilde{G}_s := V_\Sigma G_s / Z_{\text{match}}$, so that $Z_{s^{\text{phys}}} = \tilde{G}_s / g_{\text{reg}}^2$.

The susceptibility depends on $Z_{\text{match}}, V_\Sigma, G_s$ only through $\tilde{G}_s$; their separate values are pure convention. This removes the convention-dependent three-way split entirely, leaving one observable. **[DERIVED.]** It is a reparameterization, not a computation of $\tilde{G}_s$.

§7. The Canonical UV Normalization and C_{norm}

Two quantities remain, each a normalization rather than a derived number:

- $\tilde{G}_s$ (the single observable stiffness). Under the topological Kaloper–Sorbo/BF reading of the odd flux (ZS-F33), the kinetic normalization is taken canonical, $\tilde{G}_s = 1$. The physical EFT Hessian is not computed. **[ASSUMPTION S3.]**
- $c_e = 2\pi$ (the membrane-charge coefficient, $e_6 = c_e M_{\text{UV}}^2$). Large-gauge (Dirac) quantization gives c_e an $O(1)$ constant; 2π is natural, exact determination pending. **[HYPOTHESIS-strong, S4; $\approx 30\%$.]**

Canonical UV Normalization (C_{UV}): $\tilde{G}_s = 1$ and $c_e = 2\pi$.

Define $C_{\text{norm}} = \tilde{G}_s^{-1} \cdot (c_e/2\pi)^2$. Under C_{UV} , **$C_{\text{norm}} = 1$ exactly** (with no residual Z_{match} , the v1.2 inconsistency removed by folding $Z_{\text{match}}, V_\Sigma, G_s$ into $\tilde{G}_s$). $C_{\text{norm}} = 1$ is the closure gate deferred to ZS-F36. **[ASSUMPTION S5.]**

§8. The structural dimensionless factor

Assembling §§2–7 with $\hat{c}_\theta = \star$ ($v_s^2 = 6$), $Z_s^{\text{phys}} = \tilde{G}_s/g_{\text{reg}}^2$, and $e_6 = c_e M_{\text{UV}}^2$, $\chi^{(s)} = e_s^2 / (4\pi^2 Z_s^{\text{phys}}) = (v_s^2 g_{\text{reg}}^2 / 4\pi^2) \cdot \tilde{G}_s^{-1} \cdot e_6^2 = 6 g_{\text{reg}}^2 \cdot [\tilde{G}_s^{-1} (c_e/2\pi)^2] \cdot M_{\text{UV}}^4$, so that, with $6 g_{\text{reg}}^2 = 36 A/Q = (\dim Y)^2 A/Q$, $\chi^{(s)} = (\dim Y)^2 \cdot (A/Q) \cdot C_{\text{norm}} \cdot M_{\text{UV}}^4 = (1260/4807) \cdot C_{\text{norm}} \cdot M_{\text{UV}}^4$, $C_{\text{norm}} = \tilde{G}_s^{-1} (c_e/2\pi)^2$.

The arithmetic identity $36 A/Q = 1260/4807$ is **PROVEN** (an exact rational identity in the locked inputs; **I1**). That this rational number is the **physical structural factor** is **DERIVED-CONDITIONAL** on $\hat{c}_\theta = \star$ (the F34.BIV carrier reduction; §4.3). Under C_{UV} , $C_{\text{norm}} = 1$, giving $\chi^{(s)} = (1260/4807) M_{\text{UV}}^4 \approx 0.262 M_{\text{UV}}^4$, but **$C_{\text{norm}} = 1$ is not established here** — it is the F36 gate. The structural factor — what F35 derives — is the dimensionless rational $(\dim Y)^2 A/Q$.

§9. The branch ratio is a regression, not independent evidence

The single-mode value reproduces the corpus path-(c) benchmark $\chi_{83} = 0.091847 e_6^2$ via $\chi_{83}/\chi_{\text{singlet}} = 83/q_s$. **This is a tautology**: both numbers are $q \cdot g_{\text{reg}}^2/4\pi^2$, so the ratio is $83/q_s$ for **any** q_s (16.6, 13.83, 11.86 at $q_s = 5, 6, 7$) and tests nothing about the derived value. **[IDENTITY I2.]** It is a cross-version arithmetic regression, not independent evidence.

§10. The two residuals

What remains is **one dimensionless normalization gate (C_{norm}) plus one irreducible dimensionful scale (M_{UV})**. M_{UV} is the sole *dimensionful* residual, but not the sole unknown: $C_{\text{norm}} = 1$ is an undischarged dimensionless gate. By the Charge-Unit Obstruction (ZS-F33.8, **PROVEN**), M_{UV} cannot be obtained from (A, Q) and flux integrality. **[PROVEN-irreducible.]** ZS-F36 inherits both: compute $\tilde{G}_s$ and c_e (hence C_{norm}) and fix M_{UV} from one explicit UV completion.

§11. Claim ledger

#	Claim	Status	Confidence	Conditions
T1	Multiplicity-Free Duality-Singlet theorem	PROVEN (Schur)	—	unitary $\mathbb{C}$, absolutely irreducible, $d = d$, mult 1
§2	$N_{\text{eff}} = 1$	DERIVED	90%	exact- A_5 invariant-source branch
§4.1	I_h central inversion cannot swap $3 \leftrightarrow 3'$	PROVEN	—	none
§4.1	physical realization of σ	OPEN	—	inherits F34 G-Outer-Physical

#	Claim	Status	Confidence	Conditions
§4.2	★ unique traceless A_5 -invariant on the carrier	PROVEN	—	rank-6 carrier (T1)
§4.2	Lorentzian identities $\star_L^2 = -I$, $J_L = -i\star_L$, $J_L^2 = I$	PROVEN	—	none
§4.2	Euclidean $3 \oplus 3'$ norm applies to the Lorentzian EFT	DERIVED-C OND	70%	complexification/Wick identification
§4.3	$\hat{c}_\theta = \star$ (Pontryagin action + T1)	DERIVED-C OND	70%	F34.BIV carrier reduction
§4.3	$v_{s^2} = 6$ (Hilbert–Schmidt norm)	PROVEN (HS)	75% (physical)	metric norm; source identification
T2	★ non-rational on every A_5 -stable $\mathbb{Q}$ -form	PROVEN (exact)	—	none
§6	$\tilde{G}_s$ the single observable stiffness	DERIVED	90%	none
§8	36 A/Q = 1260/4807 (arithmetic)	IDENTITY / PROVEN	—	none
§8	1260/4807 is the physical structural factor	DERIVED-C OND	70%	$\hat{c}_\theta = \star$
§8	$\chi^{(s)} = (1260/4807) C_{\text{norm}}$ M_{UV}^4	DERIVED (factorization)	90%	as a factorization
§7	$C_{\text{norm}} = 1$ (absolute coefficient)	OPEN	—	C_{UV} ; F36 gate
§9	branch ratio 83/ q_s	IDENTITY	—	tautological
§10	M_{UV} the sole dimensionful residual	PROVEN-irre ducible	90%	Charge-Unit Obstruction

§12. Falsification gates

- **F-F35.1.** If $\text{End}(V_{11})$ does not decompose as stated, the construction fails. *PASS (PI).*
- **F-F35.2.** If U_+ , U_- are isomorphic, not absolutely irreducible, or unequal-dimensional, T1's commutant is not two-dimensional. *T1 hypotheses stated.*
- **F-F35.3.** If the carrier is not $\Lambda^2 V_4 = 3 \oplus 3'$ (F34.BIV), §4.2 is void.
- **F-F35.4.** If the source is not the bivector Pontryagin coupling ($\hat{c}_\theta \neq \star$), v_{s^2} changes. *Principal risk to §4.3.*
- **F-F35.5.** If $C_{\text{norm}} \neq 1$ ($\tilde{G}_s \neq 1$ or $c_e \neq 2\pi$), the absolute coefficient differs by C_{norm} ; the structural factor survives. *F36 gate.*
- **F-F35.6.** If $\star$ were rational on some A_5 -stable $\mathbb{Q}$ -form, T2 (hence the metric–arithmetic interpretation) would fail. *Excluded by the character-field proof.*

- **F-F35.7.** Any claim to fix the absolute χ_- without M_{UV} from UV data is void. $M_{UV} \rightarrow F36$.

§13. Anti-numerology

The structural factor is $(\dim Y)^2 A/Q = 36 \times 35/(437 \times 11) = 1260/4807$, all factors pre-existing locked inputs. The only integer beyond (A, Q) is $v_{-s^2} = 2d = 6$, fixed by T1 (the rank-6 carrier, $d = 3$) — not scanned. No coefficient was fitted. We **decline** to count the branch-ratio match (§9) as evidence (tautological). The disc-5 obstruction (T2) shows $v_{-s^2} = 6$ is a kinetic-norm statement, not fitted arithmetic. The *structural factor* is forced; the *absolute value* carries the unproven C_{-norm} and the UV scale M_{UV} and is not claimed.

§14. Cross-version safety

- $(A, Q, \dim Z) = (35/437, 11, 2)$, $(Z, X, Y) = (2, 3, 6)$, $g_{reg^2} = 210/4807$, $\omega = 2.2592495540$, $z^* = 0.43828 + 0.36059 i$: unchanged.
- ZS-F34's master form is not modified; v1.4 supplies a factorized value for its dimensionless content via $\tilde{G}_{-s}$ and inherits its OPEN gate G-Outer-Physical.
- ZS-F33's Charge-Unit Obstruction is preserved and used (§10).
- ZS-A19's O-A19.2 embedding question is bypassed (not closed) by the coupling route (§4.3); the unconstrained non-invariance is a NO-GO (§4.4).
- Retained from v1.0–v1.3: the single-mode reduction, embedding independence, Schur $O(\epsilon^2)$ stability. **Superseded:** the v1.1 σ -parity closure and “exact absolute coefficient”; the v1.2 C_{-norm} definition (corrected via $\tilde{G}_{-s}$ in v1.3); the v1.3 “machine-verified across” wording for T1 and its over-strong T2 universal claim; and the v1.4 verification's kron/vec transpose in the T2 generator recovery (fixed in v1.5 by a symbolic commutant, with $[s, R(g)] = 0$ now checked for all 60 g).

§15. Verification

zs_f35_verify_v1_5.py (NumPy + SymPy, seed 437) is the single source of truth (resolving the v1.3 verify/probe S3 inconsistency; T2 now fully exact via a symbolic commutant), in three honest classes:

- **PROVEN/COMPUTED (23/23 PASS):** $\text{End}(V_{11})$ decomposition (P1); arithmetic split (P2a); the character check that $\Lambda^2_{+}/\Lambda^2_{-}$ ARE 3/3' via $\chi(5A) = \phi/\phi$ (P3); the swap $R(t)\star = -\star R(t)$ with non-centrality (P4); commutant dimension 2 (P5); block-preservation (P6); the connected uniqueness $\star = \text{span of the traceless commutant}$ (P7, residual 10^{-16}); $\star$ unimodular, $\det = -1$ (P8); $v_{-s^2} = 6$ as a Hilbert–Schmidt norm (P9); the NO-GO contrast (P10); the **general theorem T1** illustrated on $A_5/S_5/S_3$ with carrier-dependence shown explicitly and the susceptibility refinement (T1-A5, T1-S5, T1-S3, T1-carrier, T1-chi); the **Lorentzian bridge** $\star_{-E^2} = +I$, $\star_{-L^2} = -I$, $J_{-L} = -i\star_{-L}$ with $J_{-L^2} = I$ (L1, L2, L3); and the **FULLY EXACT character-field obstruction T2** in $\mathbb{Q}(\sqrt{5})$, computed via a symbolic commutant (no kron/vec convention, fixing the v1.4 transpose error): integral A_5 action

and $\text{disc}(A_4) = 5$ (T2-int), exact rank 34 / nullity 2 (T2-dim), integral s with $s^2 = 5I$ and $\det = -125$ (T2-exact), $[s, R(g)] = 0$ for all $60 g \in A_5$ (T2-comm — the check that catches the transpose), and $\star = (1/\sqrt{5})s$ non-rational (T2-obs).

- **IDENTITY/REGRESSION (4/4 PASS):** 36 A/Q = 1260/4807 (I1); the branch ratio shown tautological (I2); the corrected C_{norm} (I3); the $\rho_{\Lambda, Z}$ regression (I4).
- **ASSUMPTION-CONSISTENCY (6 tracked, not derivations):** the physical split/ N_{eff} inheritance (P2b); $\hat{c}_{\theta} = \star$ (S1); $v_{s^2} = 6$ as a kinetic HS-norm, distinct from the charge lattice (S2); $\tilde{G}_s = 1$ (S3); $c_e = 2\pi$ (S4); $C_{\text{norm}} = 1$ (S5).

§16. Acknowledgements & Code Availability

This work consolidates internal Z-Spin Collaboration deep-exploration notes following ZS-F34 v1.8, incorporating five detailed peer reviews (v1.0 through v1.4). The verification script `zs_f35_verify_v1_5.py` (NumPy + SymPy) reproduces all PROVEN and IDENTITY checks, including the general theorem T1, the Lorentzian bridge, and the fully exact character-field obstruction T2; the standalone `zs_f35_T2_exact.py` reproduces the exact $\mathbb{Q}(\sqrt{5})$ computation (symbolic commutant; rank 34, $s^2 = 5I$, $[s, R(g)] = 0$ for all $60 g \in A_5$). This work used AI tools (Anthropic Claude) for verification and drafting; the author assumes full responsibility for all content, including the corrections recorded here.

§17. Appendix A — Deep-exploration (issue-tree) record

Step 0 (long list). Singlet-selection principles: (1) I_h geometric; (2) seam involution J ; (3) physical σ (S_5); (4) Pontryagin/ θ ; (5) $\star$ as the unique invariant complex structure; (6) parent-Hessian minimal mode; (7) tracelessness + A_5 -invariance on the carrier.

Generalization/strengthening: (8) group-independent theorem; (9) integral-lattice status of v_{s^2} ; (10) action-level source; (11) Lorentzian bridge; (12) exact field-of-definition obstruction.

Step 1 (MECE). Dropped (2) (distinct $\mathbb{Z}_2$, conflation rejected ZS-F12) and (6) (circular).

Retained I1–I4 for selection; promoted (8) $\rightarrow$ T1, (10) $\rightarrow$ §4.3, (11) $\rightarrow$ §4.2, (9)+(12) $\rightarrow$ T2.

Step 2–3 (tree + status). I1 = NO-GO [PROVEN]; I2 = OPEN [G-Outer-Physical]; I4 = PROVEN [T1]; I3 = DERIVED-CONDITIONAL [§4.3]. T1 PROVEN (Schur) + illustrated on three groups; the Lorentzian bridge PROVEN (L1–L3); T2 PROVEN exactly (the character field $\mathbb{Q}(\sqrt{5})$ obstructs a rational $\star$).

Step 4 (convergence). The number of OPEN nodes is one (I2, physical σ), which §§4.2–4.3 bypass. T1, T2, and the Lorentzian bridge are closed results. Converged.

Step 5 (value). The Z-Spin singlet computation is a corollary of a general theorem (T1); the lattice question becomes a genuine field-of-definition theorem (T2, exact), the most original element; the source is derived at action level (§4.3); the Euclidean–Lorentzian gap is bridged (§4.2). External value rises from an internal debt-closure to general statements about finite-symmetry top-form susceptibilities and the rational structure of duality involutions.

§18. References

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§19. Version History

- **v1.0 (June 2026)**. Initial release; parent-Hessian reduction (a reduction, not a closure). 35/35.
- **v1.1 (June 2026)**. Closure-claim revision on a flawed I_h-equivariance step. 34/34.
- **v1.2 (June 2026)**. Honest-closure revision (first review): corrected I_h ≠ S₅, carrier-uniqueness + Pontryagin, C_{UV} bundle, factorization, branch-ratio tautology, three verification classes (10+4 PASS).
- **v1.3 (June 2026)**. Surgical corrections plus generalization (second review): corrected C_{norm} via $\tilde{G}_s$; promoted to the general theorem T1; action-level $\hat{c}_\theta = \star$; the disc-5 lattice observation; status refinements; verification fixes (17+4 PASS).
- **v1.4 (June 2026)**. *Mathematical-scope and code cleanup* (third review). (1) **Restricted T1** to a finite-dimensional unitary complex representation with U₊, U₋ non-isomorphic *absolutely irreducible* of equal complex dimension d, so End_G(U_±) = $\mathbb{C}$ exactly; the norm is the *Hilbert–Schmidt* norm relative to the invariant Hermitian structure; the

susceptibility statement (iv) is refined to $\hat{c}_\theta^\dagger G^{-1} \hat{c}_\theta = |\lambda|^2 \cdot (2d)/G_J$, with the physical prefactor assembled only in the Z-Spin corollary. (2) **Resolved the verify/probe inconsistency**: a single verification script, with the carrier-dependence of $J^2 = I$ exhibited explicitly (replacing v1.3's declarative T1-rob), and the T1 wording changed from "machine-verified across" to "follows from Schur's lemma; illustrated for three groups." (3) **Upgraded T2 to an exact universal theorem**: $\text{End}_{\mathbb{Q}A_5}(3 \oplus 3') = \mathbb{Q}(\sqrt{5})$, the integral generator s with $s^2 = 5I$, and $\star = (1/\sqrt{5})s$ non-rational on every A_5 -stable $\mathbb{Q}$ -form (proven exactly in $\mathbb{Q}(\sqrt{5})$ via sympy; $\text{disc}(A_4) = 5$ is one lattice instance), replacing v1.3's over-strong " $M_{11}(\mathbb{Z})$ provably obstructed" with a precise field-of-definition obstruction. (4) **Added the Lorentzian bridge** $J_L = -i\star_L$ ($J_L^2 = I$), reinstating the v1.2 signature caveat the v1.3 draft dropped. (5) **Corrected the metric/flux-quantization framing**: $v_{s^2} = 6$ is the kinetic Hilbert–Schmidt norm; flux quantization is a separate integral-cohomology question that the Hodge involution need not respect — T2 is exactly this mismatch. (6) **Abstract** rephrased from "close the dimensionless content" to "isolate and derive the structural dimensionless factor, leaving one UV-normalization gate," and standalone normalization equations cleaned. Verification: 22 PROVEN + 4 IDENTITY (NumPy + SymPy).

- **v1.5 (June 2026, terminal). *Exact-verification and notation cleanup (fourth review)*.** (1) **Fixed the T2 verification transpose**: v1.4 recovered the commutant generator from a Kronecker system with NumPy's row-major reshape, while the vec identity is column-major, so the recovered matrix was the transpose (the theorem was unaffected, and $s^2 = 5I$ passed because it is transpose-invariant, but $[s, R(g)] = 0$ was never checked). v1.5 builds the commutant **symbolically** ($M \cdot R - R \cdot M$ with symbolic M), removing the kron/vec convention entirely. (2) **Made T2 fully exact**: the commutant dimension is now an exact sympy rank (rank 34, nullity 2), and the integral generator satisfies, exactly, $s^2 = 5I$, $\text{Tr } s = 0$, $\det s = -125$, and $[s, R(g)] = 0$ for all $60 \ g \in A_5$ (T2-comm) — the verification now matches the analytic proof. (3) **Added the general Galois-conjugate-pair form** of T2: for any such pair with real quadratic character field $\mathbb{Q}(\sqrt{m})$, the duality involution is $(1/\sqrt{m})s$ with $s^2 = mI$, never rational; A_5 ($m = 5$) is the instance. (4) **Fixed T1 notation**: replaced the inaccurate $G|_{\{V^G\}}$ with the order-parameter metric G satisfying $G(J, J) = G_{JJ}^2_{HS}$; and replaced "the identity spans the trace-carrying invariants" with the precise statement that the trace map on the 2-dimensional commutant has 1-dimensional kernel $\mathbb{C}(P_+ - P_-)$, the identity providing a canonical complementary line. (5) **Split the Lorentzian status**: the algebraic identities ($\star_L^2 = -I$, $J_L^2 = I$) are PROVEN, while applying the Euclidean $3 \oplus 3'$ norm to the Lorentzian EFT is DERIVED-CONDITIONAL on the Wick identification. (6) **Fixed the remaining §4.2 Markdown bold** on the J_L span (an underscore-adjacent-to-symbol that broke pandoc emphasis). Verification: 23 PROVEN + 4 IDENTITY (NumPy + SymPy). ($A, Q, \dim Z$) = (35/437, 11, 2) LOCKED; no new fitted parameter. **This is the**

terminal version of ZS-F35; the absolute UV normalization ($\tilde{G}_s$, c_e , M_{UV}) is the task of ZS-F36.